MATH-548 HW #1

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Problem 1.

If it is given that $x_{n+2} = x_{n+1} + x_n$ where $x_0 = 0$, $x_1 = 1$. Find the generic formula for x_n? Also $\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = ?$

Solution

Let

(1) $x_{n+2} = x_{n+1} + x_n$ &

(2) $x_0 = 0$, $x_1 = 1$.

Using the difference method for discrete problems that we covered in class,

(3) $x_{n+2} - x_{n+1} - x_n = 0$

Let's search for the solution for (3) in the following format:

(4) $x_n = e^{\lambda t}$, $x_{n+1} = \lambda e^{\lambda t}$, $x_{n+2} = \lambda^2 e^{\lambda t}$

Plugging (4) into (3) yields

(5) $\lambda^2 e^{\lambda t} - \lambda e^{\lambda t} - e^{\lambda t} = 0 \quad \Rightarrow$

as $e^{\lambda t} \neq 0$, we divide (5) by $e^{\lambda t}$ to get

(6) $\lambda^2 - \lambda - 1 = 0$

(7) $\lambda_1 = \frac{1+\sqrt{5}}{2}$ $\qquad \lambda_2 = \frac{1-\sqrt{5}}{2}$

Then,

(8) $$x_n = A\left(\frac{1+\sqrt{5}}{2}\right)^n + B\left(\frac{1-\sqrt{5}}{2}\right)^n$$

Using the initial values in (2):

$$\begin{cases} x_0 = 0 \Rightarrow x_0 = A\left(\frac{1+\sqrt{5}}{2}\right)^0 + B\left(\frac{1-\sqrt{5}}{2}\right)^0 = A+B = 0 \\ x_1 = 1 \Rightarrow x_1 = A\left(\frac{1+\sqrt{5}}{2}\right)^1 + B\left(\frac{1-\sqrt{5}}{2}\right)^1 = \frac{(A+B)+\sqrt{5}(A-B)}{2} = 1 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} A+B = 0 \\ (A+B)+\sqrt{5}(A-B) = 2 \end{cases} \Rightarrow \begin{cases} A+B = 0 \\ A-B = \frac{2}{\sqrt{5}} \end{cases} \Rightarrow$$

(9) $A = \frac{1}{\sqrt{5}}$ & $B = -\frac{1}{\sqrt{5}}$. Then (8) is

(10) $$x_n = \frac{1}{\sqrt{5}} \cdot \frac{(1+\sqrt{5})^n}{2^n} - \frac{1}{\sqrt{5}} \cdot \frac{(1-\sqrt{5})^n}{2^n}$$

$\blacksquare$

Now let's try to find $\lim_{n\to\infty} \frac{x_{n+1}}{x_n}$ using (10)

$$\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \lim_{n\to\infty} \frac{\frac{(1+\sqrt{5})^{n+1} - (1-\sqrt{5})^{n+1}}{2^{n+1}\cdot\sqrt{5}}}{\frac{(1+\sqrt{5})^{n} - (1-\sqrt{5})^{n}}{2^{n}\cdot\sqrt{5}}} =$$

$$= \lim_{n\to\infty} \frac{\cancel{2^{n}}\cdot\cancel{\sqrt{5}}\left((1+\sqrt{5})^{n+1} - (1-\sqrt{5})^{n+1}\right)}{\cancel{2^{n}}\cdot 2\cdot\cancel{\sqrt{5}}\left((1+\sqrt{5})^{n} - (1-\sqrt{5})^{n}\right)} =$$

$$= \frac{1}{2} \lim_{n\to\infty} \frac{\left((1+\sqrt{5})^{n+1} - (1-\sqrt{5})^{n+1}\right)}{\left((1+\sqrt{5})^{n} - (1-\sqrt{5})^{n}\right)} =$$

$$= \frac{1}{2} \lim_{n\to\infty} \frac{(1+\sqrt{5})^{n+1}\cdot\left(1 - \overbrace{\left(\frac{1-\sqrt{5}}{1+\sqrt{5}}\right)^{n+1}}^{\to 0 \text{ as } n\to\infty}\right)}{\cancel{(1+\sqrt{5})^{n}}\cdot\left(1 - \underbrace{\left(\frac{1-\sqrt{5}}{1+\sqrt{5}}\right)^{n}}_{\to 0 \text{ as } n\to\infty}\right)} =$$

$$\frac{1}{2} \lim_{n\to\infty} \frac{(1+\sqrt{5})\cdot 1}{1\cdot 1} = \frac{1+\sqrt{5}}{2}.$$

So $\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \frac{1+\sqrt{5}}{2}$ ← "Golden Ratio".

$\blacksquare$

Problem 2.

Given $x_{n+1} = (x_n)^3$ where $x_0 = 2$, find a generic formula for x_n?

Solution:

[I know I should use similar technique to use difference problem by defining a new variable $y_n = (x_n)^{\text{"}}$, but I could not continue till the end, so using a different method!]

Let $x_{n+1} = (x_n)^3$ & $x_0 = 2$.

Then, let's see couple initial steps;

$$x_1 = (x_0)^3 = 2^3 \quad \Rightarrow \quad n = 1$$

$$x_2 = (x_1)^3 = (2^3)^3 = 2^{3 \cdot 3} = 2^{3^2} \quad \Rightarrow \quad n = 2$$

$$x_3 = (x_2)^3 = (2^{3^2})^3 = 2^{3^2 \cdot 3} = 2^{3^3} \quad \Rightarrow \quad n = 3$$

$$x_4 = (x_3)^3 = (2^{3^3})^3 = 2^{3^3 \cdot 3} = 2^{3^4} \quad \Rightarrow \quad n = 4$$

. . . . . .

Inductive hypothesis

$$x_n = 2^{3^n}$$

For $n+1$ as an inductive step using the above hypothesis:

$$x_{n+1} = (x_n)^3 = (2^{3^n})^3 = 2^{3^n \cdot 3} = 2^{3^{n+1}}. \quad \checkmark$$

$$x_{n+1} = 2^{3^{n+1}}, \text{ then } \boxed{x_n = 2^{3^n}}.$$